

Supplementary Material

for

RADIANT-LLM: an Agentic Retrieval Augmented Generation Framework for Reliable Decision Support in Safety-Critical Nuclear Engineering

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Supplement Overview

This supplementary document accompanies the revised RADIANT-LLM manuscript. It consolidates materials moved out of the main paper to reduce manuscript length while preserving transparency and reproducibility. Specifically, it includes:

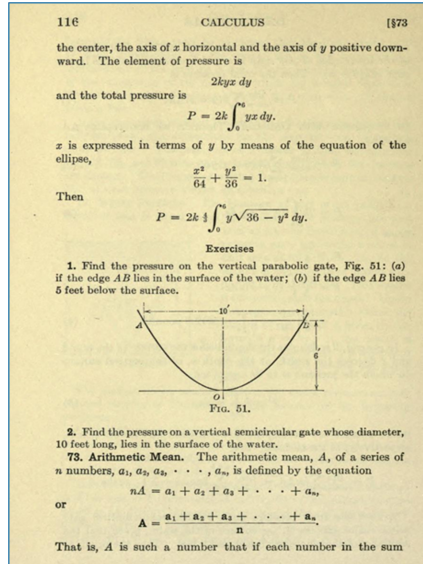
- the benchmark queries and canonical evidence anchors used for expert scoring;
- a concise summary of the scoring rubric; and
- representative workflow illustrations from the prior RADIANT-LLM study.

1 Benchmark Queries and Canonical Evidence Anchors

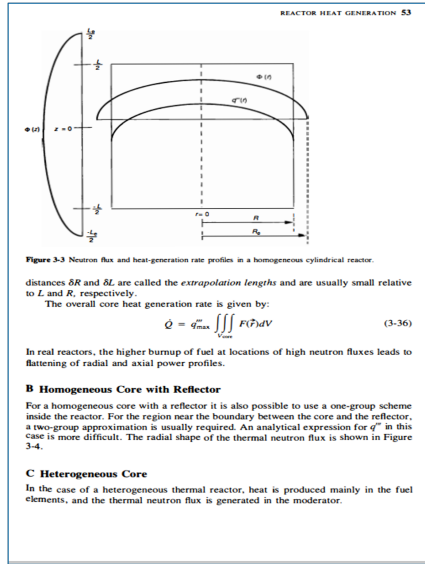
This section provides the benchmark queries and canonical evidence anchors used for expert scoring. The illustrated page-level sources are shown first, followed by the verbatim query sets for the page-level and context-scaling benchmarks.

1.1 Illustrated Source Pages for the Page-Level Benchmark

Figures [S1](#)–[S3](#) show the six sampled source pages used in the page-level benchmark, arranged as two-page composites.

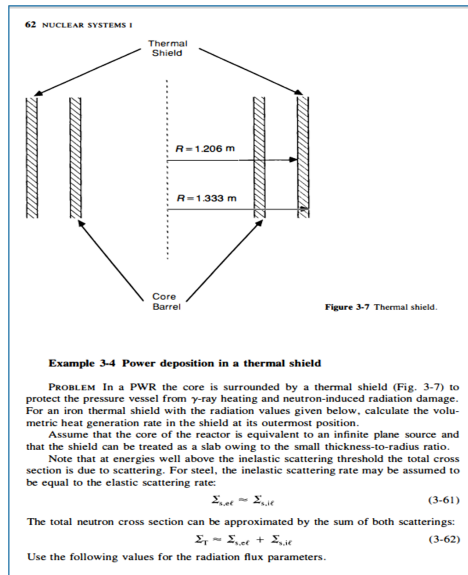


(a) Calculus Page

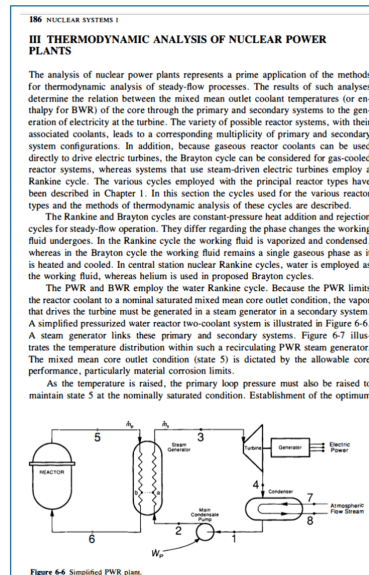


(b) Reactor Physics

Figure S1: Page-level benchmark sources: (a) gate geometry and hydrostatics; (b) cylindrical reactor flux and heat generation.

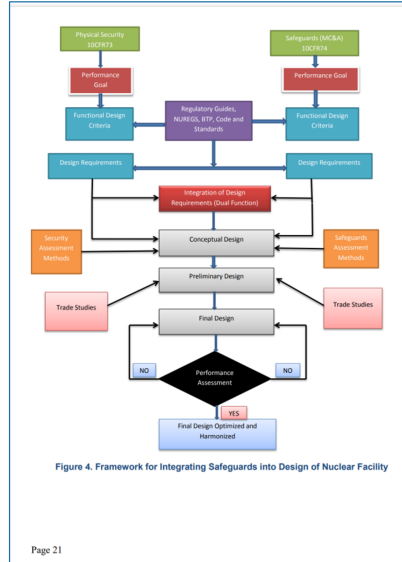


(a) Reactor Thermal Shield

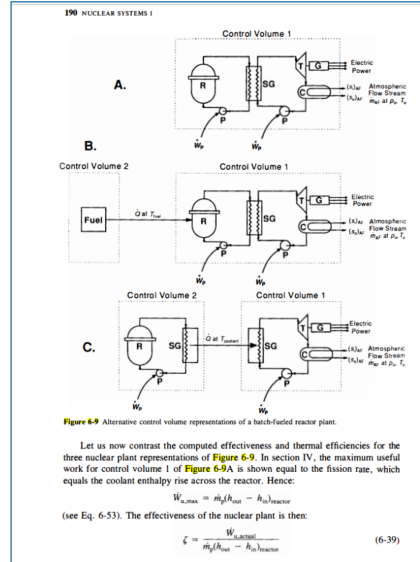


(b) Nuclear Power Plant

Figure S2: Page-level benchmark sources: (a) thermal shield geometry; (b) simplified PWR plant flow paths.



(a) UNFSF Safeguards flow chart



(b) Control volumes of a batch-fueled plant

Figure S3: Page-level benchmark sources: (a) safeguards integration workflow; (b) batch-fueled reactor control-volume representation.

1.2 Page-Level Benchmark Queries

1. **Q1.** From Fig. 51, what are the explicitly labeled geometric dimensions of the schematic (horizontal and vertical), and are these dimensions plotted axes or physical lengths?
Category: Figure 51 Dimension Extraction (ViR-dominant)
Evidence: Figure 51, Calculus_p116_73.pdf
2. **Q2.** Does the figure depict plotted x-y axes with tick marks, or a geometric schematic with physical dimensions? Justify using only visible labels.
Category: Schematic vs Axis-Plot Classification
Evidence: Figure 51, Calculus_p116_73.pdf
3. **Q3.** What ellipse equation is explicitly shown adjacent to the figure, and what semi-axis lengths does it imply?
Category: Ellipse Equation and Semi-Axis Extraction
Evidence: Figure 51 and adjacent equation text, Calculus_p116_73.pdf
4. **Q4.** What are the implied full coordinate extents of the ellipse based on the displayed equation?
Category: Ellipse Extent Derivation from Displayed Equation
Evidence: Displayed ellipse equation near Figure 51, Calculus_p116_73.pdf
5. **Q5.** According to Fig. 51 and the accompanying text, which physical quantities are to be calculated for the vertical parabolic gate, and are any geometric dimensions stated in the text that differ from those shown in the figure?
Category: Text-Figure Reconciliation for Gate Geometry
Evidence: Figure 51 and accompanying problem text, Calculus_p116_73.pdf
6. **Q6.** What spatial directions and coordinate references are explicitly labeled in Figure 3-3 for the homogeneous cylindrical reactor, and how are the radial and axial dimensions indicated?
Category: Figure 3-3 Coordinate and Marker Enumeration (ViR-dominant)
Evidence: Figure 3-3, Todreas_and_Kazimi_p53.pdf
7. **Q7.** According to Figure 3-3, how do the neutron flux and heat-generation rate profiles vary radially and axially within the homogeneous cylindrical reactor?
Category: Radial/Axial Profile Behavior Tracing
Evidence: Figure 3-3, Todreas_and_Kazimi_p53.pdf
8. **Q8.** What equation is given on this page for the overall core heat-generation rate, and what physical quantity is being integrated?
Category: Overall Heat-Generation Equation Extraction
Evidence: Equation (3-36) and surrounding text, Todreas_and_Kazimi_p53.pdf
9. **Q9.** What explanation does the text provide for the flattening of radial and axial power profiles in real reactors compared to the idealized profiles shown in Figure 3-3?
Category: Profile Flattening Explanation
Evidence: Explanatory text below Eq. (3-36), Todreas_and_Kazimi_p53.pdf
10. **Q10.** How does Figure 3-3 visually support the text's discussion of neutron flux and heat-generation behavior in a homogeneous cylindrical reactor?
Category: Figure-to-Text Support for Reactor Profiles
Evidence: Figure 3-3 and surrounding explanatory text, Todreas_and_Kazimi_p53.pdf

11. **Q11.** What radii are explicitly labeled for the thermal shield in Fig. 3-7 and what do they represent?
Category: Thermal Shield Radius Label Extraction (ViR-dominant)
Evidence: Figure 3-7, Todreas_and_Kazimir_p62.pdf
12. **Q12.** Which labeled radius corresponds to the inner surface of the thermal shield, and which corresponds to the outer surface?
Category: Inner vs Outer Radius Mapping
Evidence: Figure 3-7, Todreas_and_Kazimir_p62.pdf
13. **Q13.** Is a numerical radius explicitly given for the core barrel in Fig. 3-7?
Category: Core Barrel Radius Presence Check
Evidence: Figure 3-7, Todreas_and_Kazimir_p62.pdf
14. **Q14.** According to the text accompanying Fig. 3-7, what is the stated purpose of the thermal shield, and which physical damage mechanisms is it intended to mitigate?
Category: Thermal Shield Purpose and Damage-Mitigation Extraction
Evidence: Figure 3-7 and accompanying text, Todreas_and_Kazimir_p62.pdf
15. **Q15.** How does the text use the geometry shown in Fig. 3-7 to simplify the power-deposition calculation, and what geometric assumption is explicitly stated?
Category: Geometry-Based Simplification Assumption
Evidence: Figure 3-7 and accompanying text, Todreas_and_Kazimir_p62.pdf
16. **Q16.** According to Figure 6-6, what are the three main flow paths shown in the simplified PWR plant, and which component thermally links the two reactor-related loops?
Category: PWR Loop Identification and Coupling (ViR-dominant)
Evidence: Figure 6-6, Todreas_and_Kazimir_p186.pdf
17. **Q17.** Which labeled state points belong to the primary loop, which belong to the secondary steam-power loop, and which are associated with the condenser cooling stream in Figure 6-6?
Category: State-Point Loop Membership Mapping
Evidence: Figure 6-6, Todreas_and_Kazimir_p186.pdf
18. **Q18.** Using only Figure 6-6, trace the flow path of the secondary working fluid from the steam generator outlet back to the steam generator inlet, naming the major components and state points in order.
Category: Secondary-Loop Workflow Tracing
Evidence: Figure 6-6, Todreas_and_Kazimir_p186.pdf
19. **Q19.** What component in Figure 6-6 is directly coupled to electric-power production, and how is that relationship depicted visually?
Category: Turbine-Generator Coupling Identification
Evidence: Figure 6-6, Todreas_and_Kazimir_p186.pdf
20. **Q20.** According to the text and Figure 6-6, why does the PWR require a separate secondary system for turbine-driving vapor rather than sending primary coolant directly to the turbine?
Category: Text-Figure Justification for Separate Secondary System
Evidence: Figure 6-6 and surrounding text, Todreas_and_Kazimir_p186.pdf

21. **Q21.** What are the two top-level regulatory branches shown in Figure 4, and which CFR citations are explicitly attached to them?
Category: Figure 4 Regulatory-Branch and CFR Extraction
Evidence: Figure 4, UNFSF_p21.pdf
22. **Q22.** In Figure 4, what common sequence of design-development boxes appears under both the Physical Security and Safeguards (MC&A) branches before integration?
Category: Shared Pre-Integration Workflow Sequence
Evidence: Figure 4, UNFSF_p21.pdf
23. **Q23.** What is the exact label of the central integration box in Figure 4, and what dual-role concept does it represent in the workflow?
Category: Central Integration-Box Label and Meaning
Evidence: Figure 4, UNFSF_p21.pdf
24. **Q24.** After the Performance Assessment decision node in Figure 4, what happens on the YES path and what happens on the NO paths?
Category: Performance-Assessment Branch Logic (ViR-dominant)
Evidence: Figure 4, UNFSF_p21.pdf
25. **Q25.** Which supporting analysis activities are shown alongside the design stages in Figure 4, and at what stages do they appear?
Category: Supporting Analysis Localization by Design Stage
Evidence: Figure 4, UNFSF_p21.pdf
26. **Q26.** In Figure 6-9, how do the control-volume boundaries differ across parts A, B, and C?
Category: Control-Volume Boundary Comparison Across A/B/C
Evidence: Figure 6-9, Todreas_and_Kazimir_p190.pdf
27. **Q27.** In Figure 6-9, which part explicitly separates the Fuel from the rest of the plant, and how is the energy transfer from Fuel into the plant labeled?
Category: Fuel Separation Panel and Heat-Transfer Label
Evidence: Figure 6-9, Todreas_and_Kazimir_p190.pdf
28. **Q28.** In Figure 6-9C, which major components lie in the left control volume, which lie in the right control volume, and what heat-transfer label appears between them?
Category: Figure 6-9C Left/Right Volume Localization
Evidence: Figure 6-9C, Todreas_and_Kazimir_p190.pdf
29. **Q29.** In Figure 6-9, which plant outputs or exchanges are shown on the right side of the schematic across the control-volume representations, and how are they labeled?
Category: Right-Side Output and Exchange Label Extraction
Evidence: Figure 6-9, Todreas_and_Kazimir_p190.pdf
30. **Q30.** In Figure 6-9, which component labels appear in the plant schematic, and what do they represent in the reactor plant layout?
Category: Component-Label Interpretation in Plant Schematic
Evidence: Figure 6-9, Todreas_and_Kazimir_p190.pdf

1.3 Context-Scaling Benchmark Queries

1. **Q1.** Within the discussion of safety, security, and safeguards interfaces, which operational functions are described as being shared across all three domains, and how is this represented in the associated domain-interface schematic?
Category: Visual Overlap & Shared Functions (ViR-dominant)
Evidence: Figure 1. Safety, Security, and Safeguards Interfaces, p24
2. **Q2.** How are the roles of safety, security, and safeguards distinguished in domain-interface schematics, and what primary objective is associated with each domain as conveyed by the visual and accompanying labels?
Category: Domain Roles & Objectives (CoP-dominant)
Evidence: Figure 1. Safety, Security, and Safeguards Interfaces, p24
3. **Q3.** Which systems or operational functions facilitate interaction between the safeguards and security domains, and how are these functions presented as supporting both objectives within interface schematics?
Category: Interface Mechanisms & Dual-Use Functions
Evidence: Figure 1. Safety, Security, and Safeguards Interfaces, p24
4. **Q4.** Within the discussion of integrating safeguards and security into the design of a used nuclear fuel storage facility, which regulatory frameworks are identified as governing physical protection and material control and accounting?
Category: Regulatory Frameworks for Physical Protection and MC&A
Evidence: Section 3.1–3.2
5. **Q5.** Within the discussion of integrating safeguards and security into the design of a used nuclear fuel storage facility, what performance goal/objective is associated with each regulatory framework governing physical protection and material control and accounting?
Category: Performance Objectives of Regulatory Frameworks
Evidence: Section 3.2
6. **Q6.** What facility system is identified as having dual safeguards and security functions suitable for early integration into the design of a used nuclear fuel storage facility?
Category: Dual-Use System Identification
Evidence: Section 3.2
7. **Q7.** What criteria or regulatory considerations justify the selection of a system as having dual safeguards and security functions for early integration into the design of a used nuclear fuel storage facility?
Category: Justification for Dual-Use System Selection
Evidence: Section 3.2
8. **Q8.** Which four elements define the scope of the application of the framework for integrating safeguards and security into the design of the UNFSF?
Category: Application Scope of the Integration Framework
Evidence: Section 3 (Application of Framework)
9. **Q9.** According to the design requirements for SNM doorway monitors in the UNFSF, what minimum quantity of plutonium-239 must be detectable, under what shielding condition, and at what confidence level?

Category: SNM Doorway Monitor Detection Performance

Evidence: Section 3.3.2

10. **Q10.** What maximum false alarm rate is specified for SNM doorway monitors used to detect special nuclear material, as described in the design requirements of the UNFSF?

Category: SNM Doorway Monitor False Alarm Constraints

Evidence: Section 3.3.3

11. **Q11.** What minimum quantity of uranium-233 must a doorway monitor be capable of detecting, within what time window, and under what shielding conditions according to the specified design requirements in the design of the UNFSF?

Category: Uranium Detection Requirements and Time Constraints

Evidence: Section 3.3.2

12. **Q12.** What minimum detection effectiveness and false alarm thresholds are specified for metal and explosive detectors in the design of the UNFSF, including the minimum detectable explosive mass and required detection probability?

Category: Metal and Explosive Detector Performance Thresholds

Evidence: Section 3.3.3

13. **Q13.** In the design of the UNFSF, what happens on the YES path and what happens on the NO paths in Figure 4, after the “Performance Assessment” decision node?

Category: Figure 4 Performance-Assessment Branch Logic (ViR-dominant)

Evidence: Figure 4. 3S-by-Design Framework

14. **Q14.** In Figure 1 of the UNFSF 3S interface schematic, which callout box contains “Access Control, Detection, Monitoring, Alarms, Surveillance, MAAs, VAs, Locks & Keys,” and to what overlap region does its arrow point?

Category: Figure 1 Callout Localization and Overlap Mapping (ViR-dominant)

Evidence: Figure 1. Safety, Security, and Safeguards Interfaces, p24

15. **Q15.** In Figure 2 of the UNFSF design and regulatory process schematic, what text appears in the orange box feeding into “Design Security System,” and what is emphasized inside the green “Design Certification” oval?

Category: Figure 2 Node Extraction and Relationship Tracing (ViR-dominant)

Evidence: Figure 2. Design and Regulatory Process Schematic

16. **Q16.** In Figure 2 of the UNFSF design and regulatory process schematic, what decision step appears immediately after “Design Security System”?

Category: Figure 2 Decision-Step Tracing

Evidence: Figure 2. Design and Regulatory Process Schematic

17. **Q17.** In Figure 2 of the UNFSF design and regulatory process schematic, what two inputs are listed inside the orange box feeding into “Design Security System”?

Category: Figure 2 Input-Box Content Extraction

Evidence: Figure 2. Design and Regulatory Process Schematic

18. **Q18.** In Figure 2 of the UNFSF design and regulatory process schematic, what two items are listed under “Define Objectives”?

Category: Figure 2 Define-Objectives Extraction

Evidence: Figure 2. Design and Regulatory Process Schematic

19. **Q19.** In Figure 2 of the UNFSF design and regulatory process schematic, what is emphasized inside the “Design Certification” oval, and what is emphasized inside the “COL Application” oval?
Category: Figure 2 Oval-Emphasis Compare/Contrast
Evidence: Figure 2. Design and Regulatory Process Schematic
20. **Q20.** In Figure 2 of the UNFSF design and regulatory process schematic, how many main sequential stages appear in the central workflow, and what are they?
Category: Figure 2 Sequential-Workflow Stage Counting
Evidence: Figure 2. Design and Regulatory Process Schematic
21. **Q21.** In the UNFSF design and regulatory process schematics, which figure contains “Design Security System” and which figure contains “Design Safeguards System”?
Category: Figure 2 vs Figure 3 Central-System Label Comparison
Evidence: Figures 2 and 3. Design and Regulatory Process Schematics
22. **Q22.** In the UNFSF design and regulatory process schematics, what is emphasized inside the “Design Certification” oval in Figure 2 versus Figure 3?
Category: Figure 2 vs Figure 3 Design-Certification Comparison
Evidence: Figures 2 and 3. Design and Regulatory Process Schematics
23. **Q23.** In the UNFSF design and regulatory process schematics, what does the “COL Application” oval emphasize in Figure 2 versus Figure 3?
Category: Figure 2 vs Figure 3 COL-Application Comparison
Evidence: Figures 2 and 3. Design and Regulatory Process Schematics
24. **Q24.** In the UNFSF design and regulatory process schematics, what top-center source feeds into “Design Certification” in Figure 2, and what top-center source feeds into “Design Certification” in Figure 3?
Category: Figure 2 vs Figure 3 Top-Center Source Comparison
Evidence: Figures 2 and 3. Design and Regulatory Process Schematics
25. **Q25.** According to the UNFSF framework application text, what are the first two steps in integrating safeguards and security into the facility design?
Category: Framework-Application Initial Steps
Evidence: Section 3 (Application of Framework)
26. **Q26.** In Figure 4 of the UNFSF design integration workflow, what are the two top-level regulatory branches, and which CFR citations are explicitly attached to them?
Category: Figure 4 Top-Level Regulatory Branches
Evidence: Figure 4. 3S-by-Design Framework
27. **Q27.** In Figure 4 of the UNFSF design integration workflow, what common sequence of design-development boxes appears under both the Physical Security and Safeguards (MC&A) branches before integration?
Category: Figure 4 Common Pre-Integration Design Sequence
Evidence: Figure 4. 3S-by-Design Framework
28. **Q28.** In Figure 4 of the UNFSF design integration workflow, what is the exact label of the central integration box, and what dual-role concept does it represent?
Category: Figure 4 Dual-Function Integration Box
Evidence: Figure 4. 3S-by-Design Framework

29. **Q29.** In Figure 4 of the UNFSF design integration workflow, what design stage appears immediately after the “Integration of Design Requirements (Dual Function)” box, and what assessment step follows it?

Category: Figure 4 Post-Integration Stage and Assessment Step

Evidence: Figure 4. 3S-by-Design Framework

30. **Q30.** In Figure 4 of the UNFSF design integration workflow, which supporting analysis activities are shown alongside the design stages, and at what stages do they appear?

Category: Figure 4 Supporting Analysis Activities by Design Stage

Evidence: Figure 4. 3S-by-Design Framework

2 Scoring Rubric Summary

For completeness, this section summarizes the key rubric elements used by experts when scoring each query response. Higher values indicate better performance for CoP, CiP, CiH, and ViR, whereas lower is better for hallucination rate (HR).

- **Context Precision (CoP):** Measures correctness of the response content relative to the ground-truth points implied by the canonical evidence anchors and, where applicable, the required visual facts. Depending on the query, it may include semantic and numeric components.
- **Citation Precision (CiP):** Fraction of citations that correctly support the associated claims. Document, figure, section, or page identity must be correct and unambiguous.
- **Citation Hit (CiH):** Binary indicator of whether at least one citation hits a canonical evidence anchor defined for the benchmark item.
- **Hallucination Rate (HR):** Fraction of generated atomic claims that are unsupported by the retrieved evidence.
- **Visual Recall (ViR):** Fraction of required visual facts correctly recovered in the response for visually grounded items.

3 Representative Workflow Illustrations from the Prior RADIANT-LLM Study

These figures are included to preserve a compact visual record of the earlier interactive CSV-to-PDF-to-QA workflow without occupying main-manuscript space.

127.0.0.1:54900/radiant_llm

RADIANT-LLM

Illuminating Pathways to Secure & Safe Nuclear Design

Settings

Select a Model (*)

gpt-4.1-augmented

Initialize

'gpt-4.1' model initialized successfully.

Working Directory (Optional)

C:\

redacted

Using working directory:

C:\

Redacted

Get to know radiant_llm-AI

Select a query(optional)

Conversation

Great job! Now, invoke the PDD reader tool and process these documents. Study them and prepare for 3S for nuclear-related questions.

Submit

Query: From the CSV data in this directory, find 5 documents where the 'DESCRIPTION' column contains '3s' and list their 'OSTI_IDENTIFIER' and 'URL' in a tabular format here.

RadiantLLM:

Here are 5 documents from your CSV data where the DESCRIPTION column contains "3s," along with their OSTI_IDENTIFIER and URL:

OSTI_IDENTIFIER	URL
2455109	https://www.osti.gov/biblio/2455109
1167010	https://www.osti.gov/biblio/1167010
1592894	https://www.osti.gov/biblio/1592894
1226427	https://www.osti.gov/biblio/1226427
2403492	https://www.osti.gov/biblio/2403492

If you need further details or wish to analyze the content of any of these documents, please let me know!

Query: Now, for each of these URLs, invoke the File downloader tool and download these PDFs from the OSTI.gov website into this working directory.

RadiantLLM:

Here are the results of attempting to download the requested PDFs from OSTI.gov into your specified working directory:

OSTI_IDENTIFIER	Download Status	Local Filename	Direct PDF Link
2455109	Failed	N/A	https://www.osti.gov/servlets/purl/2455109
1167010	Failed	N/A	https://www.osti.gov/servlets/purl/1167010
1592894	Success	1592894.pdf	https://www.osti.gov/servlets/purl/1592894
1226427	Success	1226427.pdf	https://www.osti.gov/servlets/purl/1226427
2403492	Success	2403492.pdf	https://www.osti.gov/servlets/purl/2403492

- The downloads for OSTI identifiers 1592894, 1226427, and 2403492 were successful.
- The downloads for 2455109 and 1167010 failed due to a remote connection issue from the OSTI server.

If you need further processing or analysis of the successfully downloaded PDFs, please specify your requirements!

Figure S4: Web-based GUI interaction showing the RADIANT-LLM agent querying OSTI meta-data, identifying nuclear 3S-relevant records, downloading selected PDFs from approved URLs, and preparing the documents for downstream RAG processing.

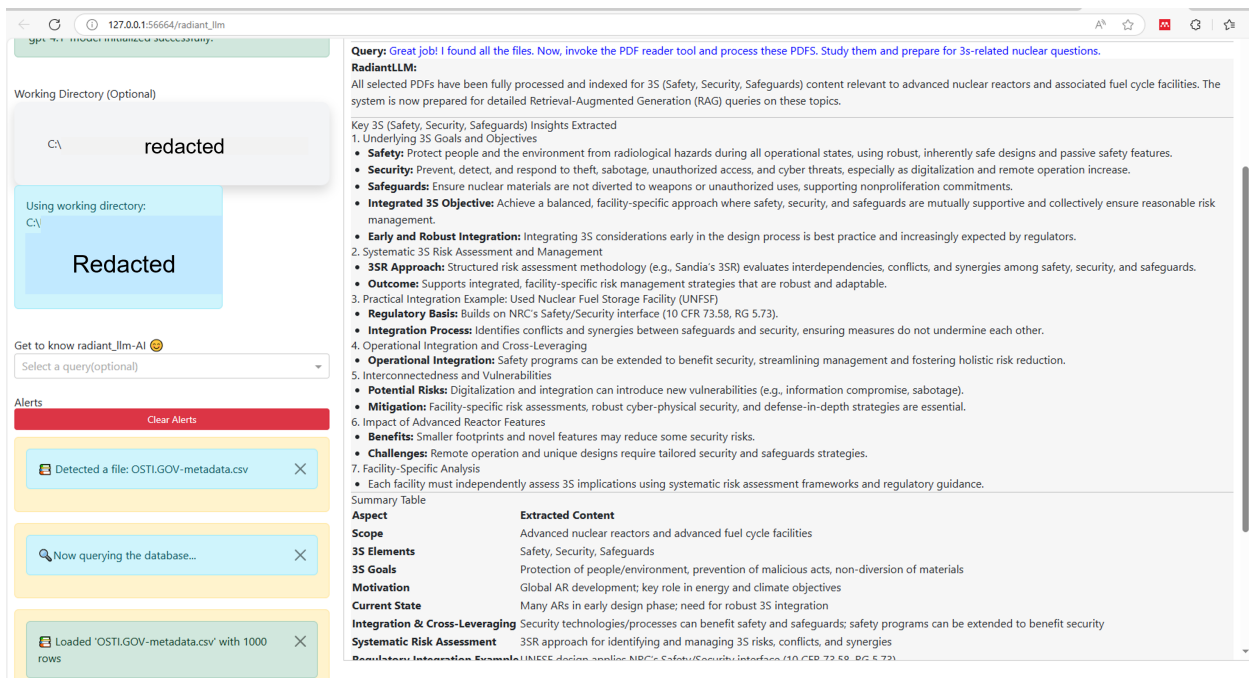


Figure S5: Web-based GUI interaction showing the RADIANT-LLM agent invoking the PDF reader workflow after document download, processing and indexing the selected PDFs, extracting nuclear 3S-relevant insights, and preparing the resulting knowledge base for document-grounded question answering.